MIL-L-36460A
12 December 1973
SUPERSEDING
MIL-L-36460
6 December 1965

MILITARY SPECIFICATION

LENS, OPHTHALMIC, NEUTRAL ABSORPTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers neutral absorption ophthalmic lenses suitable for use in correcting defective vision.

1.2 Classification. The lenses shall be of the following types, classes and styles, as specified.

Type I - Lens blanks.
Type II- Lens, single vision, uncut.

Class A - Bifocal, semifinished.
Class B - Semispheres.
Class C - Simple.
Class D - Compound.

Style 1 - Spheres.
Style 2 - Plus plano cylinders.
Style 3 - Plus on plus compounds.
Style 4 - Minus on plus compounds.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup, Paperboard.

FSC 6540

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Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INCORPORATED

ANSI Z80.1-1972	American National Standard Requirements for First-Quality Prescription Ophthalmic Lenses.

(Application for copies should be addressed to the American National Standards Institute, Incorporated, 1430 Broadway, New York, N. Y. 10018).

3. REQUIREMENTS

3.1 Material.

3.1.1 Lens. The lens shall be made of high grade neutral absorption, ophthalmic quality crown glass with an index of refraction of 1.5230, $\pm$ 0.0015. The lens shall conform to the transmission requirements specified herein. It shall be durable and of suitable composition and hardness to give good service in use, resist the action of weather, and not be easily scratched or damaged under ordinary conditions. The glass shall be free of bubbles, inclusions, and striae.

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3.1.1.1 Transmittance.

3.1.1.1.1 Daylight transmittance. The daylight transmittance through 2.2 mm ± 0.2mm, of glass shall be within the limits given in table I. Daylight transmittance shall be computed as shown in table II.

3.1.1.1.2 Near-infrared transmittance. The near-infrared transmittance through 2.2 mm ± 0.2 mm shall be as specified in table I when computed as the average spectral transmittance at 50 millimicron steps in the spectral region of 800 to 1,200 millimicrons.

3.1.1.1.3 Erythemal ultraviolet transmittance. The erythemal ultraviolet transmittance through 2.2 mm ± 0.2 mm shall be as specified in table II, when computed as the average spectral transmittance at wavelengths of 290, 300, 310 and 320 millimicrons.

3.1.1.1.4 Chromaticity coordinates. The chromaticity coordinates, x and y, of the lens shall be limited to the trapezoid designated by the following points, (FIGURE 1)based on purity from 6 to 14 percent, a dominant wavelength of 570 to 580 millimicrons, and a thickness of 2.2 mm. Coordinates shall be computed as shown in table II.

x = .3182
y = .3305

x = .3288
y = .3495

x = .3384
y = .3402

x = .3223
y = .3264

3.1.1.1.5 Deviation in spectral transmittance. The spectral transmittance of the lenses may vary with wavelengths between 430 and 730 millimicrons, in order that the average percentage deviation within 9 spectral bands from daylight transmittance is less than 12 when computed as shown in table III, and when tested as specified in 4.4.6.

Table I. Limiting values of transmittance

	Percent Transmittance		
Neutral lenses	Near infrared	Erythemal ultraviolet	Daylight
Maximum	15 (but always less than daylight)	0.2	18
Minimum	0	0	12

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Table II Computation table of coordinates for "Illuminant C"

Wave length (millimicrons)	xP	yP	zP	T	xPT	yPT	zPT
380	4		20	0.104	0	0	2
390	19		89	0.240	5	0	21
400	85	2	404	0.301	26	1	122
410	329	9	1,570	0.275	90	2	432
420	1,238	37	5,949	0.174	215	6	1,035
430	2,997	122	14,628	0.114	342	14	1,668
440	3,975	262	19,938	0.118	469	31	2,353
450	3,915	443	20,638	0.127	497	56	2,621
460	3,362	694	19,299	0.137	461	95	2,644
470	2,272	1,058	14,972	0.142	323	150	2,126
480	1,112	1,618	9,461	0.144	160	233	1,362
490	363	2,358	5,274	0.145	53	342	765
500	52	3,401	2,864	0.147	8	500	421
510	89	4,833	1,520	0.149	13	720	226
520	576	6,462	712	0.151	87	976	108
530	1,523	7,934	388	0.153	233	1,214	59
540	2,785	9,149	195	0.154	428	1,409	30
550	4,282	9,832	86	0.155	663	1,524	13
560	5,880	9,841	39	0.157	923	1,545	6
570	7,322	9,147	20	0.158	1,157	1,445	3
580	8,417	7,992	16	0.159	1,338	1,271	3
590	8,984	6,627	10	0.160	1,437	1,060	2
600	8,949	5,316	7	0.160	1,432	855	1
610	8,325	4,176	2	0.160	1,332	668	0
620	7,070	3,153	2	0.161	1,138	509	0
630	5,309	2,190		0.161	855	353	0
640	3,693	1,443		0.160	591	233	0
650	2,349	886		0.159	373	141	0
660	1,361	504		0.159	216	80	0
670	708	259		0.158	112	41	0
680	369	134		0.157	58	21	0
690	171	62		0.156	27	10	0
700	82	29		0.153	13	4	0
710	39	14		0.151	6	2	0
720	19	6		0.149	3	1	0
730	8	3		0.148	1	0	0
				Totals	15,085 (X)	15,512 (Y)	16,023 (Z)

Notes

(a) $X = \sum x\,PT$; $Y = \sum y\,PT$; $Z = \sum z\,PT$

(b) Daylight transmittance, Tc = Y/1,000 = 15.5 percent

(c) Chromaticity corrdinates;

$x = X/(X + Y + Z) = 15,085/46,620 = 0.324$

$y = Y/(X + Y + Z) = 15,512/46,620 = 0.333$

(d) Daylight transmittance, Tc, and chromaticity coordinates, x any y, for standard illuminant C.

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(e) The symbol "T" presents spectral transmittance, the ratio of transmitted to homogeneous radiant flux.
(f) Tables II and III are worksheets for determining the coordinates and the daylight duplication index of a typical neutral glass.

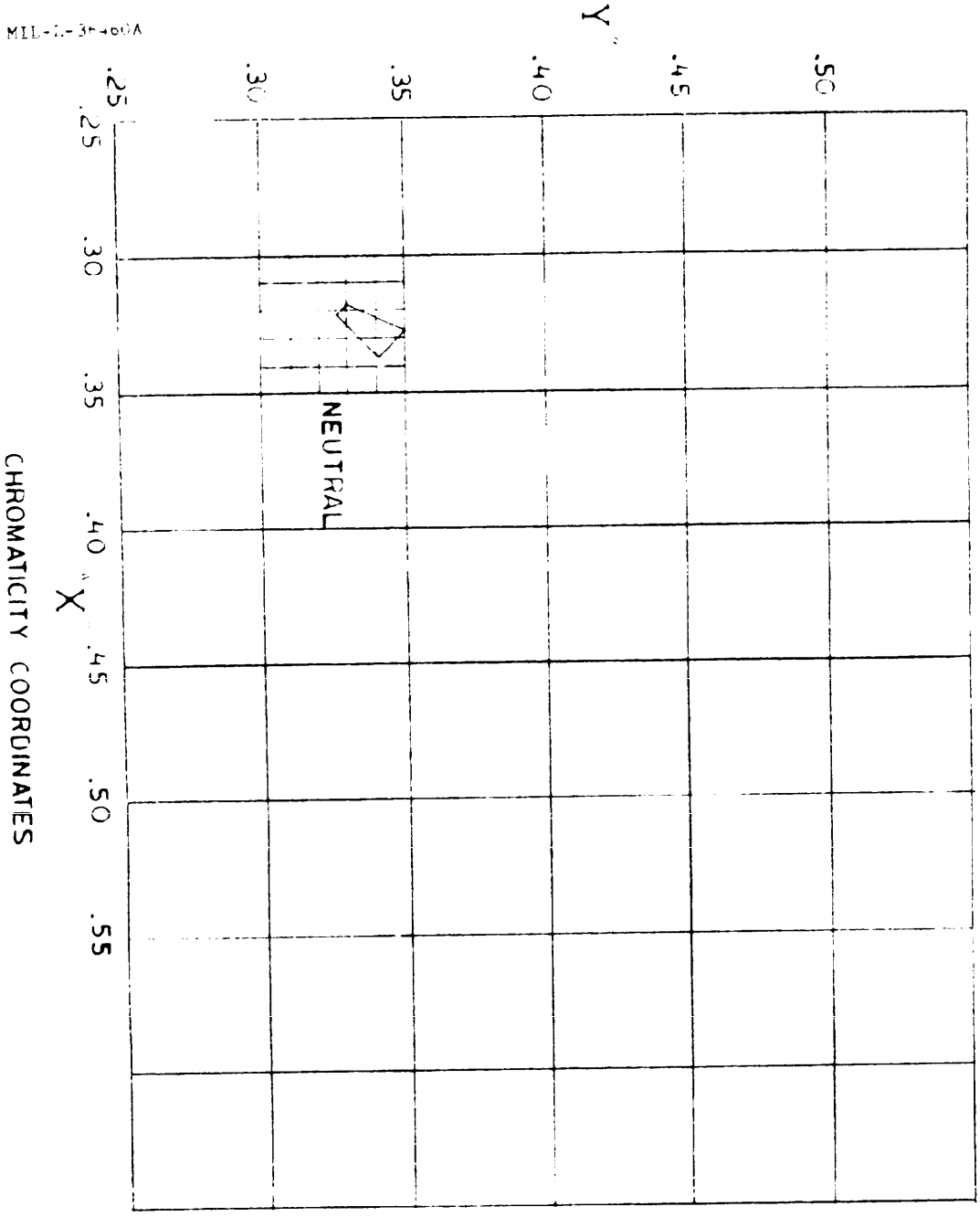
Table III Deviations of spectral transmittance

Wave length (millimicrons)	T	Band n	Wave length range	Average transmittance Tn	Percent deviation 100(1-Tn/Tc)	Weight	Product
430	0.114						
440	0.118						
450	0.127						
460	0.137	1	430-490	0.133	14	5	70
470	0.142						
480	0.144						
490	0.145	2	460-520	0.145	7	10	70
500	0.147						
510	0.149						
520	0.151	3	490-550	0.151	3	10	30
530	0.153						
540	0.154						
550	0.155	4	520-580	0.155	0	10	0
560	0.157						
570	0.158						
580	0.159	5	550-610	0.159	3	10	20
590	0.160						
600	0.160						
610	0.160	6	580-640	0.160	3	10	30
620	0.161						
630	0.161						
640	0.160	7	610-670	0.160	3	10	30
650	0.159						
660	0.159						
670	0.158	8	640-700	0.158	2	5	10
680	0.157						
690	0.156						
700	0.153	9	670-730	0.153	1	1	1
710	0.151						
720	0.149				Totals	71	261
730	0.148						

Notes

(a) Daylight transmittance, Tc = 0.155.
(b) T = Transmittance at 10 micromicrons band
(c) Tn = Average transmittance of 60 micromicrons band

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CHROMATICITY COORDINATES

FIGURE 1

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Notes (con't)

(d) The average transmittance Tn for a given band is the average of the seven tabulated values within that band, except that the first and last values are divided by 2, and the average computed by dividing the sum of the values by 6. (Average deviation = 261/71 = 3.7 percent)
(e) Tables II and III are worksheets for determining the coordinates and the daylight duplication index of a typical neutral glass.

3.1.2 Segment. Where applicable, the segment of the lens shall be made of glass of the type commonly known in the optical industry as barium. It shall be free of bubbles, inclusions, striae, and cloudiness. The segment shall not be tinted. The dispersion of the segment glass, Nu Value (v), shall be equal to or greater than the value defined by the formula:

$$v \text{ equals } \frac{16.7\ (n-1)}{n - 1.3744}$$

Where "n" is the index of refraction of the segment glass for the sodium D line.

3.2 Construction.

3.2.1 Type I - Lens blanks.

3.2.1.1 Class A - Bifocal, semifinished.

3.2.1.1.1 Major blank with fused segment. The semifinished lens blank shall be fabricated of the material specified in 3.1.1 and shall have a fused straight top bifocal segment (see 3.2.1.1.2). The blank shall be free of bubbles or other defects which might result from the fusing process. The bifocal segment shall be of the addition specified (see 6.2), within a tolerance of ± 0.09 diopter (see 6.3.4 for definition). The addition power shall be measured as specified in 4.4.1. The distance from the geometric center of the major blank to the top of the segment shall be 6 mm, ± 0.5 mm. Unless otherwise specified, the segment shall be decentered to the right or left 3.0 mm, plus or minus 1.0 mm. The segment side of the blank shall be ground and polished to a spherical base curve of 6.00 diopters, plus or minus 0.06 diopter. The front surface shall be free from any localized refractive error due to any defect which impairs the image quality, when the questioned area is examined either visually or with a standard lens measuring instrument such as a Vertometer or Lensometer. The thickness of the blank shall be 6 mm, ± 0.5 mm.

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3.2.1.1.2 Segment. The segment shall be of the material specified in 3.1.2 and shall be of the straight top type. The segment diameter (measured at the widest portion) shall be 25 mm, ± 0.5 mm. However, when the lens blanks are paired, the paired segment diameters shall be equal, within a tolerance of 0.5 mm. The mechanical center of the segment shall be 5 mm, ± 0.5 mm below the flattop.

3.2.1.2 Class B - Semisphere. The semisphere lens shall be molded in meniscus form, from the material specified in 3.1.1. The convex surface shall be ground and polished on a 6.00 diopter, ± 0.06 diopter spherical base curve. The concave surface shall be rough, fine ground, or molded surface with a curve of approximately 6 diopters. The thickness of the lens shall be 5.5 mm, ± 0.5 mm.

3.2.2 Type II - Lens, single vision, uncut.

3.2.2.1 Class C- Simple. The lens shall be made of the material specified in 3.1.1.

3.2.2.1.1 Style 1 - Spheres. The sphere shall be single vision uncut, simple lens with plus or minus spherical power. The axial power shall correspond with plus or minus spherical, dioptric power specified, with allowable tolerances (3.2.2.3.1). The marginal power shall correspond with axial spherical and cylindrical power, as specified, with allowable tolerances (3.2.2.3.2).

3.2.2.2 Class D- Compound. The lens shall be made of the material specified in 3.1.1.

3.2.2.2.1 Style 2 - Plus plano cylinders. The single vision, uncut lens shall be plano cylinder with the cylinder curve located on the convex surface. The axial power shall correspond with plus cylindrical power specified, with allowable tolerances (3.2.2.3.1). The marginal power shall correspond with axial spherical and cylindrical power, as specified, with allowable tolerances (3.2.2.3.2).

3.2.2.2.2 Style 3 - Plus on plus compounds. The single vision uncut, plus on plus compound lens shall have the cylinder curve located on the convex surface. Axial power shall correspond with plus spherical and plus cylindrical power as specified, with allowable tolerance (3.2.2.3.1). The marginal power shall correspond with axial and cylindrical power as specified, with allowable tolerance (3.2.2.3.2).

3.2.2.2.3 Style 4 - Minus on plus compounds. The single vision, uncut, minus on plus compound lens shall have the cylinder curve located on the convex surface. The axial power shall correspond with minus spherical and marginal power shall correspond with axial spherical and cylindrical power as specified, with allowable tolerances (3.2.2.3.2).

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3.2.2.3 Tolerances.

3.2.2.3.1 Axial power. The refractive power measured through the optical center shall agree to the specified power to within the following tolerances:

Powers zero through 6.00	± 0.06 diopters
Powers 6.00 through 12.00	± 1 percent of power
Powers above 12.00	± 0.12 diopters

The axial power shall be measured as specified in 4.4.3.1.

3.2.2.3.2 Marginal power. The tolerances for marginal power shall be as follows:

(1) The marginal meridional power referred to the vertex sphere at points A and B (see Figure 2) shall agree to the specified power within the tolerances for meridional power shown in tables 2 through 6 of American National Standards Institute (ANSI) Standard Z80.0-1972.

(2) The cylindrical power, referred to the vertex sphere, at points A and B (see Figure 2) shall agree to the specified power within the tolerances for cylindrical power shown in tables 2 through 6 of ANSI Standard Z80.1-1972. The marginal power shall be measured as specified in 4.4.3.2.

3.2.2.4 Thickness. The center thickness of Type II lenses shall be a minimum of 2.0 mm. Thickness at a distance of 1.5 mm from any edge shall be a minimum of 1.0 mm. for all lenses.

3.3 Sizes. Unless otherwise specified the lenses shall be round, 63 mm, ± 0.5 mm in diameter.

3.4 Surface characteristics. The polished surfaces of the lens shall be free from surface waves, scratches, or pits. The lens surface shall be of such quality that they appear black against a black background, with side illumination. The lens surface shall be tested as specified in 4.4.2.

3.5 Workmanship. Workmanship shall be first class throughout. Ophthalmic lenses shall be free from striae, bubbles, chips, or other defects which would impair their ophthalmic quality.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor. Lot identity shall be maintained and shall be demonstrated to the Government.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality shall be zero for all sample sizes. Unit of product for test purposes shall be the lens.

Table IV. Sampling plan

For examination	Inspection level	AQL	
		Major	Minor
Surface power, thickness, segment size and location.	S-2	1.0	
Lens size, surface defects internal defects.	II	1.0	2.5
For tests			
Refractive power and addition power	S-3	*	

[illegible] in conformance with [illegible]

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Table IV. Sampling plan (cont'd)

For examination	Inspection level	AQL	
		Major	Minor
Daylight, near-infrared and erythemal transmittance, Chromaticity coordinates Deviation in spectral transmittance.	S-2	*	

* Acceptance number shall be zero.

4.3 Examination. The lens shall be examined for defects including, but not limited to those listed in table V.

Table V. Classification of defects

Category	Defect
Major	
101	Surface power out of tolerance.
102	Defects (Pits, scratches, bubbles, straie or inclusions - located over 3 mm from any edge).
103	Thickness not as specified.
104	Segment size out of tolerance.
105	Segment location not as specified.
106	Wave distortion.
107	Cylinder curve on concave side.
Minor	
201	Size below minimum.
202	Cloudiness or haze.
203	Jagged and sharp edges.

4.3.1 Physical quality and appearance. Lenses shall be inspected against a dark background, in light from an open-shaded, 40 watt incandescent daylight lamp, with the lens 12 inches from the light source. No pits, scratches, water-marks, bubbles, striae, or inclusions shall be acceptable. Minute hairline scratches shall not be a cause for rejection. Waves found by visual inspection shall be passable if no deterioration in image quality is found when the localized area is examined with a standard lens measuring instrument.

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4.3.1.1 Other Glass defects. Lenses shall be examined for glass defects and other irregularities affecting the optical qualities, by moving the lens in front of the aperture of a lens measuring instrument. Sudden jumps or blurring of the image will indicate a glass defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Bifocal addition power. Using a recognized standard lens measuring instrument which gives a measure of vertex power, the bifocal addition power shall be obtained using the following procedure:

> Place the segment side of the bifocal toward the light source, and take the difference between the distance and reading portion powers. The measurement through the distance portion must be made as far above the optical center of the distance portion as the measurement through the segment is below the optical center of the distance portion. The target images should be centered by the use of auxiliary prisms. The bifocal addition power shall meet the requirements specified in 3.2.1.1.1.

4.4.2 Surface quality. Lenses shall be tested for surface quality by examining them with side illumination of not less than 100 foot candles against a black background. A lens with satisfactory quality will appear black. Grayness will indicate an unsatisfactory surface quality.

4.4.3 Methods of measurement.

4.4.3.1 Axial powers. The axial powers may be measured on any standard lens measuring instrument capable of yielding the required accuracy.

4.4.3.2 Marginal powers. The marginal powers may be measured on any standard lens measuring instrument that has been appropriately modified to yield the required accuracy referred to the vertex sphere.

One modification of standard instrument that is satisfactory, is to add a hemispherical surface (see figure 2) to act as the seating surface against which the lens under test is held. The radius of the surface is 27 mm. This surface is so arranged that the edge of the standard opening is tangent to the surface of the sphere. A dot is marked on the sphere to locate a point 30 degrees from the axis. The optical center of the lens is pressed against this dot during the test. Suitable prisms are used to bring the image of the tangent within the field of view of the observing telescope. When the measurements are made in this manner, the observed values are the values referred to the vertex sphere and no corrections are necessary.

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4.4.4 Surface power. The surface power shall be measured with an instrument such as a lens measure, a lens clock or Geneva gage.

4.4.5 Thickness measurement. Thickness shall be measured using a standard lens caliper or similar suitable measuring instrument.

4.4.6 Spectral transmittance. The lenses shall be measured for spectral transmittance by a spectrophotometer, or equal. Each melt shall be tested to determine conformance with spectral transmittance requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. Two lenses (1 pair), of the stock number specified, constitute one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV, shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table VI. Packaging Quantities

Type	Class	Style	Lens Power	Type of Unit Package	No. of Units (Pairs) to be packaged in each intermediate Carton	No. of Units (Pairs) to be packed in each Exterior Container
I	A	-		Carton	5	180
I	B	-		Envelope	5	180
II	C	1	0.00 to 2.50	Envelope	20	720
II	C	1	2.25 to 2.75	Envelope	15	540
II	D	2	0.00 to 1.50	Envelope	20	720
II	D	3	0.25 to 1.25	Envelope	20	720
II	D	4	0.25 to 1.50	Envelope	20	720
II	D	4	1.25 to 2.75	Envelope	15	540

NOTE: Quantities indicated in Intermediate Carton Column and Exterior Container Column in Table VI are mandatory, except when the number of units indicated will not fit the standard intermediate carton, a variation of -5 only, for classes C and D is permitted. Quantities indicated in Exterior Container Column will be adjusted accordingly to maintain the standard of 36 intermediate cartons per Exterior Container.

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5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit packaging. Type 1, class A lens blanks shall be individually enclosed in tissue or other suitable material to protect the lens from damage, and further packaged in an individual carton of appropriate size and design. Closure of carton shall be adequate to prevent spillage of contents under normal handling. Two individual cartons (right and left) placed together constitute a matched pair. All other lens blanks and lenses shall be packaged in individual envelopes of appropriate size and design.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-676, except that commercial colors shall be acceptable. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Special requirements as specified in Table II of PPP-B-636 shall apply for fiberboard boxes.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

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5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. In addition, fiberboard boxes shall conform to the special requirements specified in Table I of PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual lens carton or envelope. Each individual carton shall be marked with base curve, addition of the segment, diameter of the segment, right or left eye, and name or registered trade-mark of the contractor. Each envelope shall be marked with power, type, center thickness, and name or registered trade-mark of the contractor.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The neutral absorption lenses are used as sunglasses in aviators style metal frame and worn by pilots, observers and aircrew personnel.

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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Type, class, style, size and power required (see 1.2)

(c) Base curve and addition power required (if applicable).

(d) Selection of applicable levels of packing and packing.

6.3 Definitions.

6.3.1 Vertex sphere. The vertex sphere (see figure 2) is a spherical surface with a radius of 27 mm, and tangent to the rear surface of the lens at its axial point.

6.3.2 Marginal points. The marginal points A and B (see figure 2), are chosen for measurement of marginal power. Points A and B lie on the axis of maximum and minimum power, respectively, and together with the optical center subtent angles of 30^{0} at the center of the vertex sphere.

6.3.3 Corrected curve lens. A corrected curve lens shall be one member of an integrated series of ophthalmic lenses having surface curvatures especially computed and charted, which reduce the visually important monochromatic aberrations to the least significant feasible in commercial manufacture, for at least a 60 degree vista measured on the eye side, for specified centers of rotation and object distances.

6.3.4 Power of an addition. The power of an addition is defined as the difference between the powers measured through the distance and segment portions of the blank when the segment side is placed against the stop of of a recognized standard lens measuring instrument.

6.4 This specification does not include all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or which are commercially indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project Number 6540-1032

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A. GIVE PARAGRAPH NUMBER AND WORDING

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